

Desigualdade de Cauchy-Schwarz

Se $a_1, a_2, \dots, a_n$ e $b_1, b_2, \dots, b_n$ são números reais arbitrários.

Então vale a seguinte desigualdade

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right) \dots \dots (*)$$

O sinal de igualdade é válido, se e somente se, existe um número $\dots (*)$ real $\lambda \in \mathbb{R}$, tal que $a_k \lambda + b_k = 0$, para todo $k (= 1, 2, \dots, n)$.

Demonstração.

$$f(x) = \sum_{k=1}^n \underbrace{(a_k x + b_k)^2}_{\text{elevado ao quadrado}} \geq 0$$

$$f(x) = \sum_{k=1}^n (a_k x + b_k)^2 = Ax^2 + 2Bx + C \dots \dots (1)$$

$$\text{sendo } A = \sum_{k=1}^n a_k^2, \quad B = \sum_{k=1}^n a_k b_k, \quad C = \sum_{k=1}^n b_k^2$$

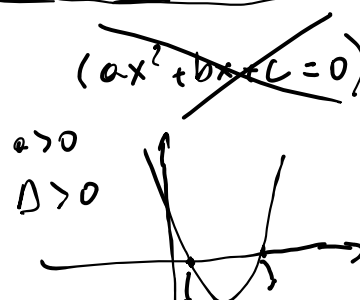
Se $n=2$, então $f(x) = \sum_{k=1}^2 (a_k x + b_k)^2$ resumo

$$\begin{aligned} \therefore f(x) &= (a_1 x + b_1)^2 + (a_2 x + b_2)^2 \\ &= a_1^2 x^2 + 2b_1 a_1 x + b_1^2 + a_2^2 x^2 + 2a_2 b_2 x + b_2^2 \\ &= \underbrace{(a_1^2 + a_2^2)}_{\equiv A} x^2 + \underbrace{2(a_1 b_1 + a_2 b_2)}_{\equiv B} x + \underbrace{(b_1^2 + b_2^2)}_{\equiv C} \\ &= Ax^2 + 2Bx + C \end{aligned}$$

$$Ax^2 + 2Bx + C \geq 0$$

$$\Delta = 4B^2 - 4AC \leq 0$$

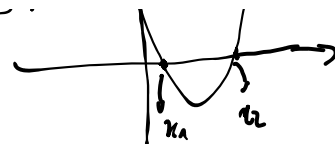
$$B^2 - AC \leq 0$$



$$B^2 - AC \leq 0$$

$$\Rightarrow \boxed{B^2 \leq AC}$$

$$B = \sum_{k=1}^n a_k b_k, A = \sum_{k=1}^n a_k^2, C = \sum_{k=1}^n b_k^2$$



$$"\Rightarrow" \exists x_0 \in \mathbb{R} \mid a_k x_0 + b_k = 0, \forall k=1, 2, \dots, n \Rightarrow B^2 = AC$$

$$f(x) = \sum_{k=1}^n (a_k x + b_k)^2 \geq 0$$

$$f(x_0) = \sum_{k=1}^n (a_k x_0 + b_k)^2 = 0, \quad x_0 = -\frac{b_k}{a_k}$$

$$f(x) = \sum_{k=1}^n (a_k x + b_k)^2 = Ax^2 + 2Bx + C$$

$$0 = f(x_0) = \sum_{k=1}^n (a_k x_0 + b_k)^2 = Ax_0^2 + 2Bx_0 + C = 0$$

$$f(x_0) = A \left(x_0 + \frac{B}{A} \right)^2 - \frac{B^2}{A} + C = 0, \quad x_0 = -\frac{B}{A}$$

$$\boxed{\begin{aligned} A \left(x + \frac{B}{A} \right)^2 - \frac{B^2}{A} + C &= Ax^2 + 2Bx + \frac{B^2}{A} - \frac{B^2}{A} + C \\ &= Ax^2 + 2Bx + C \end{aligned}}$$

$$f(x_0) = A \left(-\frac{B}{A} + \frac{B}{A} \right)^2 + C - \frac{B^2}{A} = 0$$

$$\therefore C - \frac{B^2}{A} = 0 \Rightarrow \boxed{B^2 = AC}$$

$$f(x) = Ax^2 + 2Bx + C = 0$$

$$\Delta \geq 0, \quad \Delta = 4B^2 - 4AC \geq 0 \Rightarrow B^2 \geq AC$$

$$AC \leq B^2 \leq AC \Rightarrow B^2 = AC$$

$$AC \leq b^2 \leq AC \Rightarrow B^2 = AC$$

" $\Leftarrow$ " $B^2 = AC$, então $\exists x_0 \in \mathbb{R} \mid a_k x_0 + b_k = 0, \forall k \in \mathbb{N}$

$$f(x) = Ax^2 + 2Bx + C = \sum_{k=1}^n (a_k x + b_k)^2 = 0$$

$$\Delta = 4B^2 - 4AC = 4(B^2 - AC) = 0$$

$$\Delta = 0 \Rightarrow \exists x_0 \mid f(x_0) = 0 \leftarrow$$

$$f(x_0) = \sum_{k=1}^n \underbrace{(a_k x_0 + b_k)}_{=0}^2 = 0$$

$$f(x_0) = \sum_{k=1}^n \underbrace{(a_k x_0 + b_k)^2}_{\geq 0} \geq 0$$

$$f(x_0) = 0 \text{ se e somente se } a_k x_0 + b_k = 0$$

$$\Rightarrow x_0 = -\frac{b_k}{a_k}$$

